

NET24N

Universal control panel for 24V operators Operating instructions and warnings

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SYMBOLS

The following symbols are used within this manual to indicate potential hazards.

	Important safety notice. Failure to follow these instructions may result in serious injury or damage to property. Failure to do so may result in product malfunction and create a hazardous situation.
	Important safety notice. Contact with live parts can result in death or serious injury.
	Important information for installing, programming or commissioning the product.

1 PRODUCT DESCRIPTION

NET24N is a universal control panel for **DEA** System 1 or 2 24V --- operators automations with or without encoder.

The main feature of this control board is its ease of configuration of inputs and outputs according to any needs thus ensuring adaptability to any type of automation. It is therefore easy to set up and exclude all unnecessary functions.

2 TECHNICAL DATA

Power supply (V)	230 - 240 V ~ (50/60 Hz)	
Rated power transformer (VA)	See wiring diagram	
Fuse F1 (A) (transformer)		
Fuse F2 (A) (batteries input)	Fuse T 15A L 250V	
Outputs 24V  motors <u>maximum</u> output current (A)	2 x 7A (oppure 1 x 10A)	
	Warning: In absolute terms, the maximum current for each output should not exceed 10A when using a single motor and 7A when using 2 motors.	
Auxiliaries power supply output	24V 	(24V_AUX + 24V_ST = max 200mA)
Stabilized power supply output for safety devices	24V 	
“Warning” output	24V  max 15W	
Electric lock output	max 1 art. 110 or 24V  max 5W configurable output	
Flashing light output	24V  max 15W	
Operating temperature range (°C)	-20÷50 °C	
Receiver frequency	433,92 MHz	
Transmitters type of coding	HCS fix-code - HCS rolling code - Dip-switch - DART	
Max remote controllers managed	100	

3 ELECTRICAL CONNECTIONS



! Risk of injury and material damages due to electric shock !



! Risk of malfunction due to improper installation !

Make the connections by following the indications of the wiring diagram.

WARNING For adequate electrical safety, all cables must have double insulation. Be sure to keep them clearly separated (**minimum 4 mm in air or 1 mm through additional insulation**) safety extra-low-voltage cables from low-voltage cables (230V ~) by arranging them within plastic raceways and securing them with appropriate clamps near the terminal blocks.

WARNING For the connection to the network, use multipolar cable of minimum section 3x1,5mm² and the type required by current regulations. To connect the motors, use cable minimum section 1.5 mm² and the type required by current regulations. By way of example, if the cable is outside (outdoors), it must be at least equal to H05RN-F while, if inside (in channel), it must be at least equal to H05VV-F.

WARNING All cables should be stripped and unsheathed in the immediate vicinity of the terminals. Keep the cables slightly longer so that any excess can be removed later.

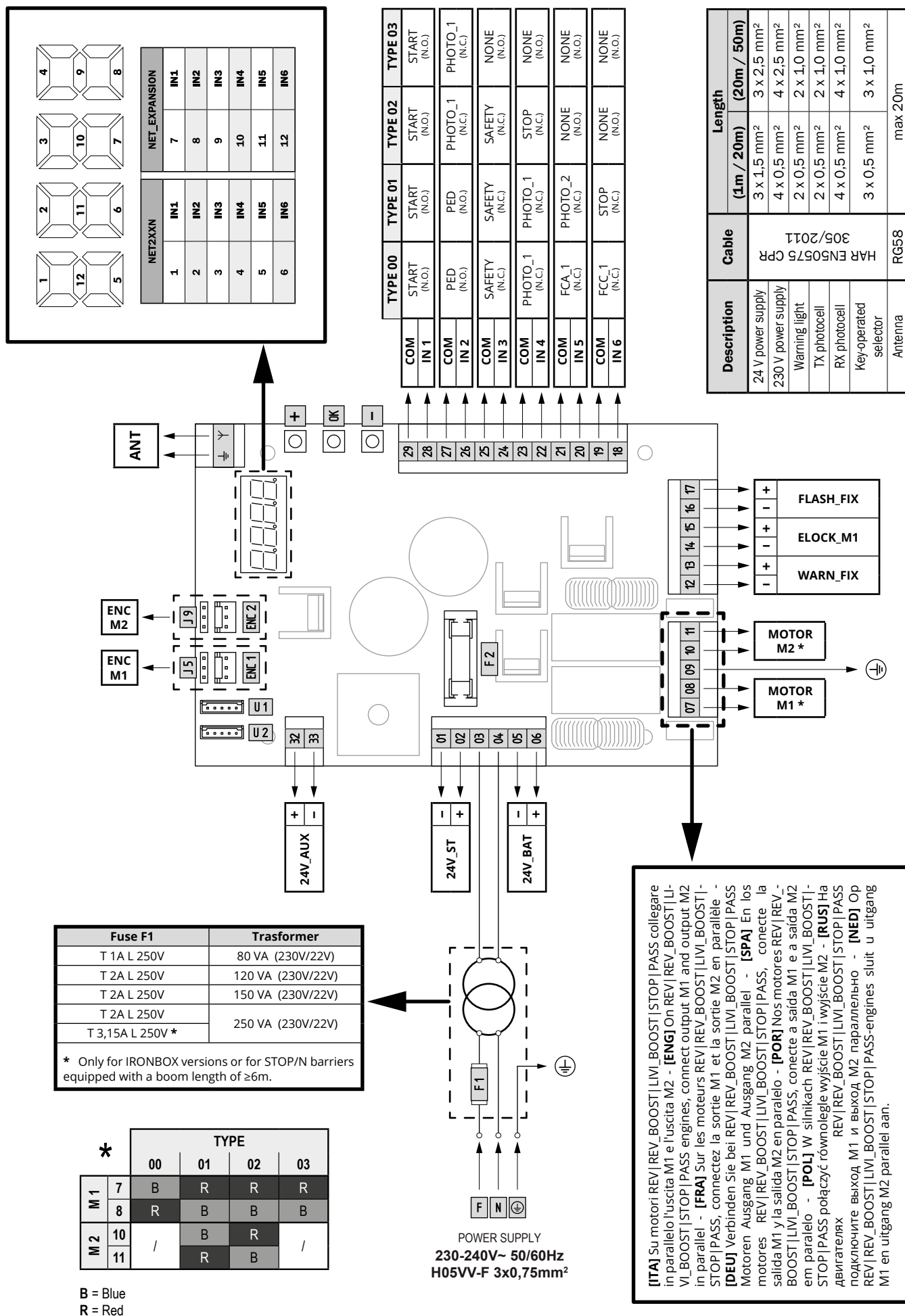
WARNING Connect the grounding conductor to the appropriate terminal, taking care to keep its length longer than that of the active conductors so that in the event of the cable coming out of the fixing seat, the active conductors are the first to stretch.

WARNING When connecting the encoder to the control board, use only a dedicated 3x0,22mm² cable.

NET24N board terminal table

3 - 4		22 V ~ transformer power supply input or photovoltaic accumulator GREEN ENERGY input.		
5 - 6		24 V === battery power supply		
7 - 8		Motor Output 1: 24V max 7A (max 10A if only one motor is present)		
9		Connection of motors metallic parts		
10 - 11		Motor Output 2: 24V max 7A (if present)		
12 - 13		24 V === max 15 W output for open gate fix warning light (if P052=0), flashing (if P052=1) or courtesy light (if P052>1)		
14	-	“BOOST” output for electric-lock, max 1 x art. 110 (if P062=0), 24V pulse output, max 5W (if P062=1), step by step (if P062=2), electro-brake output for not self-locking operators (if P062=3), output for electric-lock power supply via external relay (if P062=4), output for electro-magnets power supply for barriers (ifP062=5) or temporized output (if P062>5).		If the installation requires different commands and / or additional to the standard, you can configure each input to the required rate. Refer to Chapter “Advanced Programming”.
15	+			
16 - 17		24 V === Flashing light output max 15W		
18	IN 6	INPUT 6 Configurable Input (See P022 for selectable values)		
19	Com			
20	IN 5	INPUT 5 Configurable Input (See P021 for selectable values)		
21	Com			
22	IN 4	INPUT 4 Configurable Input (See P020 for selectable values)		
23	Com			
24	IN 3	INPUT 3 Configurable Input (See P019 for selectable values)		
25	Com			
26	IN 2	INPUT 2 Configurable Input (See P018 for selectable values)		
27	Com			
28	IN 1	INPUT 1 Configurable Input (See P017 for selectable values)		
29	Com			
ANT	⌋	Radio antenna signal input		
	⌋	Radio antenna ground input		
32	+	24 V === power supply output for auxiliary devices		(24V_AUX + 24V_ST) = max 200mA
33	-			
1	-	Stabilized 24 V === power supply output for tested safety devices		
2	+			
F1		See wiring diagram		
F2		Fuse T 15A L 250V		
ENC_M1		M1 motor encoder input	Encoder selection Jumper (J5=M1 - J9=M2) Position “  A” = operators with encoder (remind to set P029=0) Position “  B” = operators without encoder (remind to set P029=1)	
ENC_M2		M2 motor encoder input		
U 1		UART 1 Plug-in module input for NET-NODE and MEMONET		
U 2		UART 2 Plug-in module input for NET-EXP		

WIRING DIAGRAM



4 CONFIGURATION OF THE CONTROL PANEL

The universal control unit NET24N can be used for the management of the following types (**TYPE**) of closures motorized by **DEA** System: swing and sliding gates, overhead doors and barriers.

In order to ensure maximum adaptability to each **TYPE** of closure, the control board provides an initial procedure, performed only at the first turn, for the optimal configuration of inputs, outputs and parameters (see diagram **A**). Once configured, the control panel will operate in the mode "dedicated" to the **TYPE** of selected closing. After performing the initial configuration it is sufficient to execute the standard programming for the installation on which it is operating.

All settings remain in memory even in the case of subsequent flare-ups (see diagram **B**).

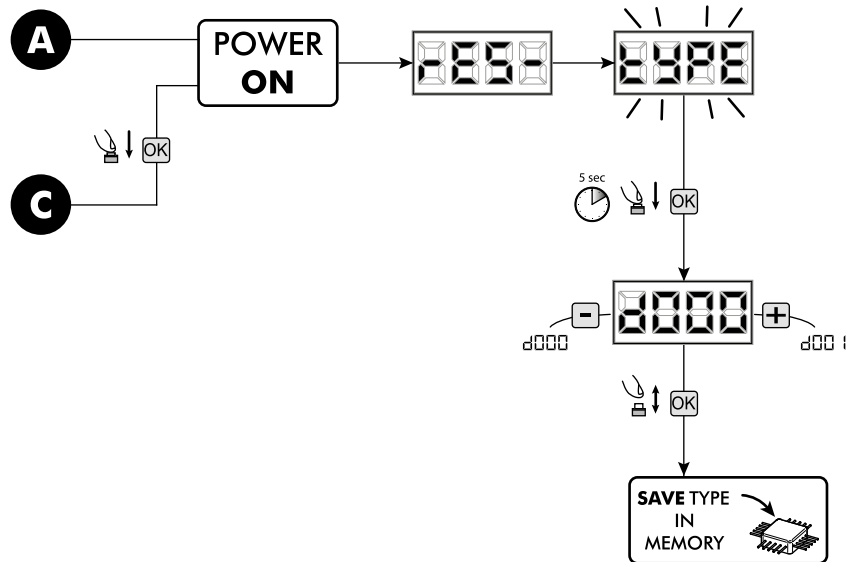
If necessary the **TYPE** of configured closing can be later adjusted following diagram **C**.

FIRST CONTROL BOARD IGNITION

Configuration after the first ignition

A For the first control panel ignition, proceed as follows:

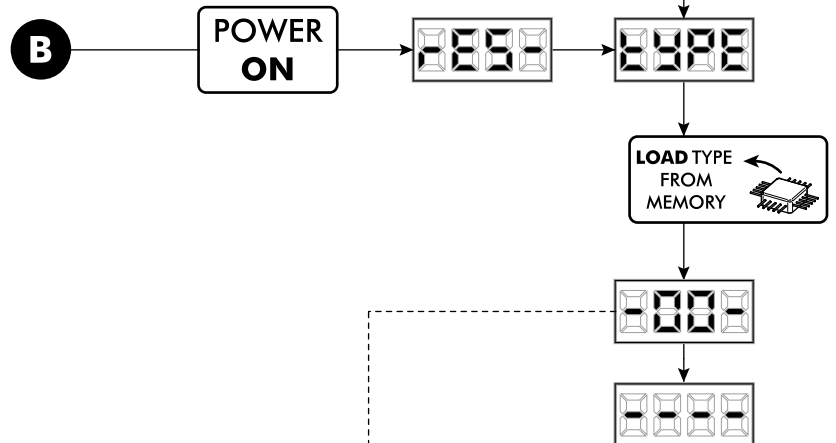
1. Apply power, the display shows in sequence the writing "rES-" and "TYPE" flashing;
2. Press the **[OK]** button and hold for 5 seconds until the display shows "d000" on the display;
3. Acting on the **[+]** and **[-]** keys, select the desired configuration depending on the type of installation (es. d002) and confirm by pressing the **[OK]** button;
At this point, the selection will be stored and reloaded each time in the future.
4. Follow signs, "TYPE", "-00-" followed by the symbol of closed gate "----".



Following ignitions

B If you have already saved a configuration, proceed as follows:

Apply power, the display shows in sequence the writing "rES-", "TYPE", "-00-" followed by the symbol of closed gate "----".



Modify the existing configuration

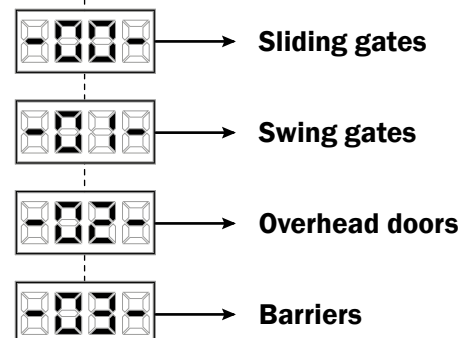
C If you have already saved a configuration and you want to change it, proceed as follows:

1. Hold down the **[OK]** button and give power, the display shows in sequence the writing "rES-" and "TYPE" flashing;
2. Press the **[OK]** button and hold for 5 seconds until the display shows "d000" (the value changes to match the previous configuration used) on the display;
3. Acting on the **[+]** and **[-]**, select the new desired configuration depending on the type of installation (es. d002) and confirm by pressing the **[OK]** button;

NOTE: Stop the reconfiguration procedure prior to confirmation, involves loading the previous configuration by the control panel without any modification.

NOTE: However, if the reconfiguration procedure is brought to an end, the new configuration will take the place of the previous one and will be reloaded each time in the future.

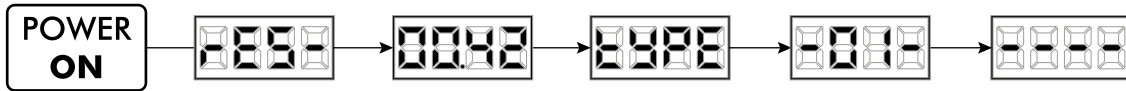
4. Follow signs, "TYPE", "-00-" followed by the symbol of closed gate "----".



5 STANDARD PROGRAMMING

1 Power Supply

When turned on, “rES-”, “00.42” (or the current firmware version) “TYPE”, “-01-” (or the selected Type) appear on the display in sequence followed by the closed gate symbol “- - -”.



* If the control panel has already been programmed and the power fails or is switched off - once power is returned and a START command is given, the position reset procedure is performed (see “rESP” in the table “WORKING STATUS MESSAGES” on page 37).

2 Visualisation of inputs and operations-counter status

1. Scroll the parameters with the [+] and [-] keys until the screen reads “P013”;
2. Access the parameter by pressing the [OK] button;
3. The “Input Status” is shown on the screen (check that this is correct):

☐ OPEN CONTACT ☒ CLOSE CONTACT

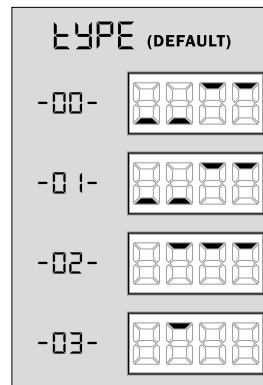
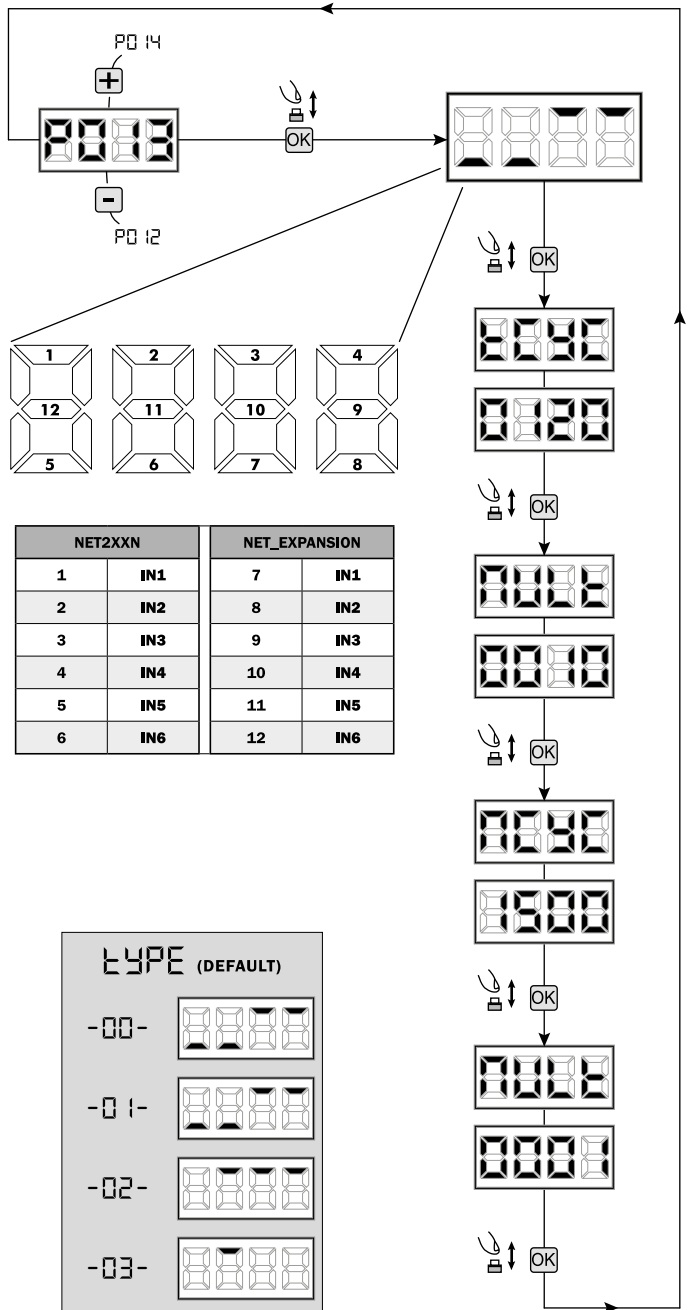
4. Press the [OK] button again;
5. The “Total Operation Counter” “tCYC” appears on the screen followed by the “MULT” multiplier. To calculate the number of completed operations, the two values must be multiplied.

I.e.: “tCYC” = 120x10 = 1200 operations completed

6. Press the [OK] button again;
7. The “Total Maintenance Counter” “MCYC” appears on the screen followed by the “MULT” multiplier. To calculate the number of operations remaining before the maintenance request, the two values must be multiplied.

I.e.: “MCYC” = 1500x1 = 1500 operations yet to be completed before the maintenance request

8. Press the [OK] button again to exit the parameters (P013 is shown on the screen again).



3 Selection type of operators

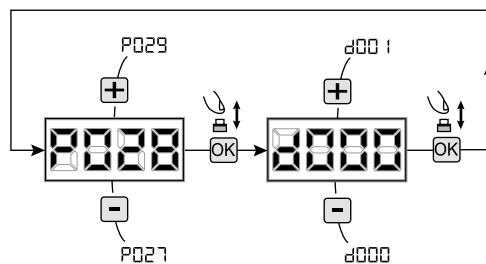
! IMPORTANT !

1. Scroll down the parameters with **[+]** and **[-]** keys until you visualise P028;
2. Access the parameter by pressing the **[OK]** key;
3. Acting on **[+]** and **[-]** keys, set:

Type 00	Type 01	Type 02	Type 03
• 005 6/24N (BOOST) - 6/24X (BOOST) • 006 9/24N - 9/24X • 007 REV24 • 008 REV24/M BOOST	• 000 GEKO • 001 LOOK - MAC - STING • 002 GHOST 100 - GHOST 200 • 003 502/24 - ANGLO • 004 502MT/24 • 005 GEKO/X	• 003 902/24 - 905/24 • 004 902R/24	• 003 PASS 24/N • 004 STOP 24/N

Warning: If you are using non **DEA** System operators, set the parameter on the closer value for family type and performances.

4. Confirm your choice by pressing the **[OK]** key (display returns again to P028).



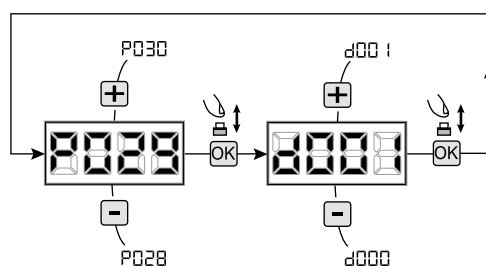
4 Selection operating with or without encoder

! IMPORTANT !

Warning: Remember to correctly set the jumpers J5 and J9.

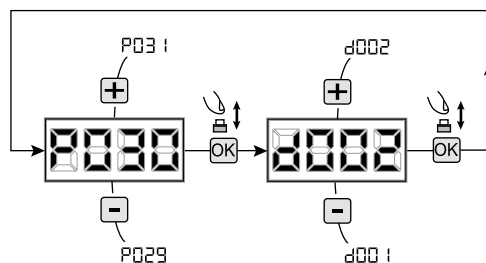
- A**
☒ A position = operators with encoder (remind to set P029=0)
- B**
☐ B position = operators without encoder (remind to set P029=1)

1. Scroll down the parameters with **[+]** and **[-]** keys until you visualise P029;
2. Access the parameter by pressing the **[OK]** key;
3. Acting on **[+]** and **[-]** keys, set:
 - d000=for operators with encoder;
 - d001=for operators without encoder;
4. Confirm your choice by pressing the **[OK]** key (display returns again to P029).



5 Selection 1 or 2 operators functioning

1. Scroll down the parameters with **[+]** and **[-]** keys until you visualise P030;
2. Access the parameter by pressing the **[OK]** key;
3. Acting on **[+]** and **[-]** keys, set:
 - d001=for a single motor operating;
 - d002=for 2 motors operating;
4. Confirm your choice by pressing the **[OK]** key (display returns again to P030).

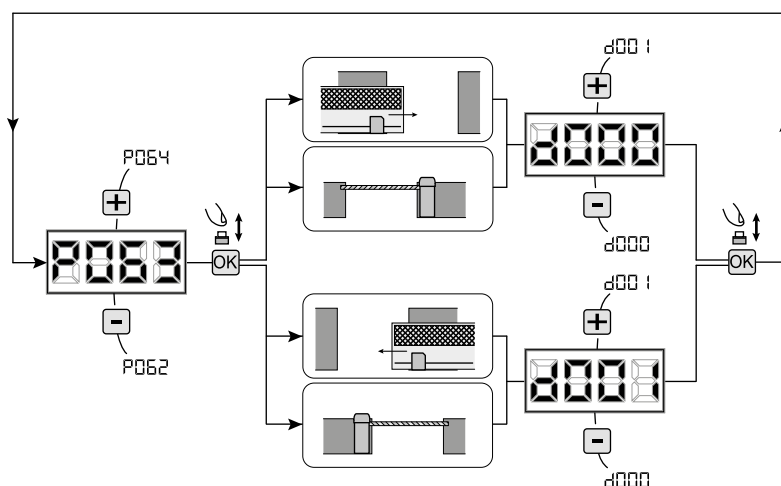


6 Selection of direction of motion (only Type 00 and Type 03)

1. Scroll down the parameters with **[+]** and **[-]** keys until you visualise P063;
2. Access the parameter by pressing the **[OK]** key;
3. Acting on **[+]** and **[-]** keys, set:
 - d000=motor in standard position;
 - d001=motor in inverted position;
4. Confirm your choice by pressing the **[OK]** key (display returns again to P063).

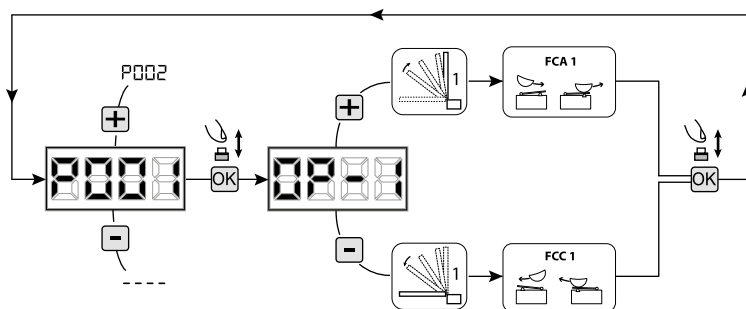
Warning: The parameter automatically reverses the motors output open/close.

Warning: Changing this parameter you need to change the parameters for the opening and closing limit switches.



7 How to adjust the limit switch

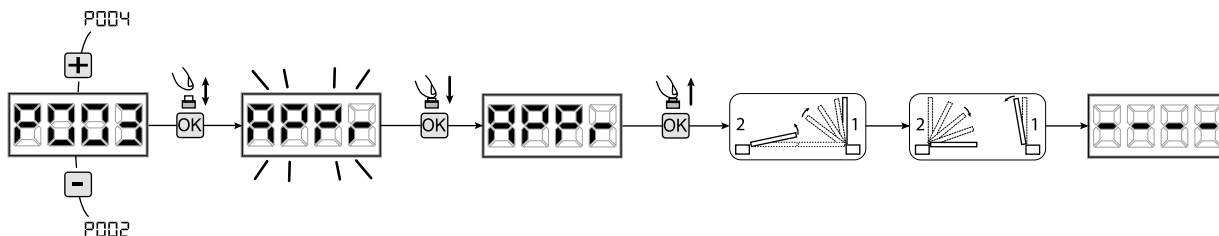
1. Scroll down the parameters until you visualize P001;
2. confirm by pressing the **[OK]** key;
3. by pressing **[+]** (**OPEN**) and **[-]** (**CLOSE**), move the leaf in the opening position and adjust the limit switch cam so that it pushes the microswitch in that point; Repeat adjusting the closing limit switch.
4. Confirm by pressing the **[OK]** key (display shows again P001).



WARNING If the Operator 2 is present, repeat the previous settings using P002.

8 Motor stroke learning

1. Scroll down the parameters with **[+]** and **[-]** keys until you visualise P003;
2. Access the parameter by pressing the **[OK]** key;
3. When "**APPr**" flashes, continue pressing the **[OK]** key;
4. Release the **[OK]** key when "**APPr**" stops flashing; Start the learning procedure with operator 1 opening (if it starts closing, disconnect the power supply, inverse the operator cables and repeat the operation);
5. Wait for the door (or doors in case of using 2 motors) searches and stops on the opening stop and then on the closing stop.
If you want to anticipate the stopping strokes in opening, you can manually intervene by giving an impulse to "Start" button (or pressing the "OK" on the control panel) simulating the stroke.
6. Once the procedure is ended, the display will show "----".

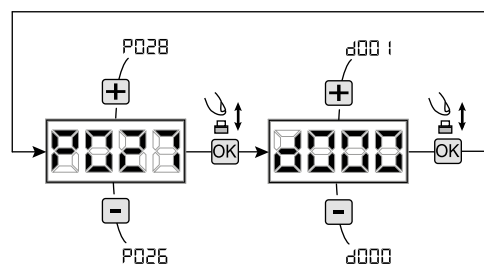


WARNING (only Type 01 and Type 03) Once you have executed the learning stroke, operate a complete cycle (opening/closing) and then check the manual release to make sure it is working properly. If it's to "hard" increase the value of P057 of 1 or more.

9 Transmitters learning

9.1 Transmitters coding selection

1. Scroll down the parameters with **[+]** and **[-]** keys until you visualise P027;
2. Confirm by pressing on the **[OK]** key;
3. Select the type of transmitter by scrolling **[+]** and **[-]** keys:
 - d000=fix rolling-code (**suggested**);
 - d001=complete rolling-code;
 - d002=dip-switch;
 - d003=DART;
4. Confirm by pressing on the **[OK]** key (display shows again P027).

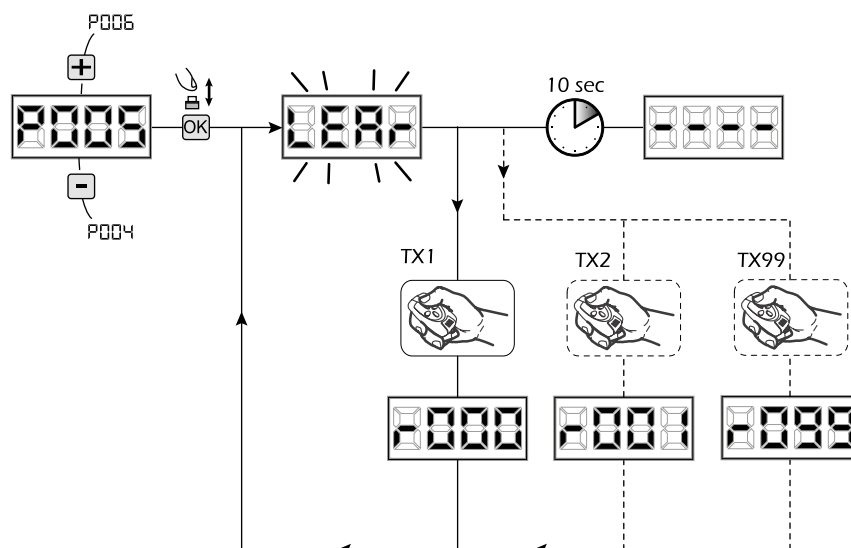


Warning: If you need to vary the type of encoding, and only if other remotes with different encoding are memorized, you need to erase memory (P004)

AFTER you have set the new encoding.

9.2 Learning

1. Scroll down the parameters with **[+]** and **[-]** keys until you visualise P005;
2. Confirm by pressing on the **[OK]** key;
3. When the symbol “LEAr” appears, press on any key of the transmitter you want to memorize;
4. The display visualizes the number of the transmitter just memorized and then “LEAr”;
5. Memorize all necessary transmitters repeating this procedure from step 3;
6. Wait 10 seconds before quitting the memorization mode, display shows now “----”.



Warning: In the case of rolling code remotes, the receiver can be put into learning mode by pressing the hidden button on a remote control previously learned.

Warning: When using personalized transmitters, after entering P005 the learning of the first personalized transmitter is possible only by pressing its hidden button. Afterwards, only transmitters personalized with the same encryption key can be memorized (through the usual procedure), unless a memory reset is carried out (P004).

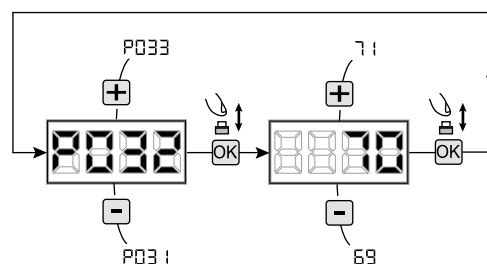
Warning: If poor radio range is experienced, it is recommended to connect the antenna of the flashing light (if present) or to install a tuned external antenna.

10 Adjustment of operating parameters

If you need to modify the operating parameters (force, speedness etc..):

1. Scroll down the parameters until you visualize the desire parameter (i.g. P032);
2. Confirm by pressing on the **[OK]** key;
3. By pressing on **[+]** and **[-]**, set up the desired value;
4. Confirm by pressing on the **[OK]** key (display shows the parameters previously selected).

For the complete list of the “Operating Parameters” See the table on page. 38.



11 Programming complete

WARNING At the end of the programming procedure, use the buttons **[+]** and **[-]** until the appearance of the symbol “---”, the operator is now ready again for new manoeuvres.

To perform any “Advanced Programming” operations (cancellation of the remotes, configuration inputs, etc. ..), see on page 33.

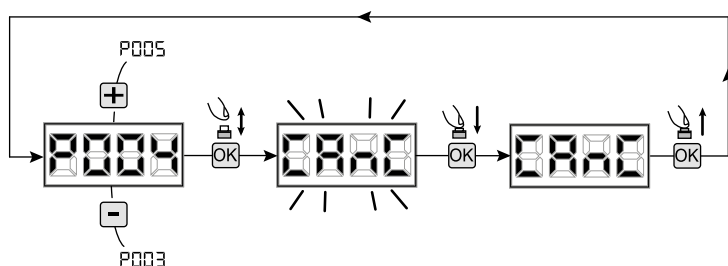
6 ADVANCED PROGRAMMING

Here are some added programming procedures relating to remotes memory management and advanced configuration of the control inputs.

1 Deletion of memorized transmitters

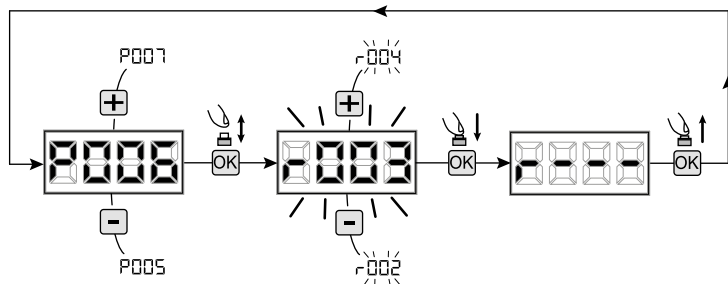
1.1 Deletion of all transmitters

1. Scroll down the parameters until you visualize P004;
2. Confirm by pressing on the **[OK]** key;
3. When “**CAnC**” is flashing, press the **[OK]** key for a few seconds;
4. Release the **[OK]** key as soon as “**CAnC**” stops flashing;
5. All memorized transmitters have been deleted (display shows again P004).



1.2 How to search and delete a transmitter

1. Scroll down the parameters until you visualize P006;
2. Confirm by pressing on the **[OK]** key;
3. By pressing on **[+]** and **[-]**, keys, select the transmitter you want to delete (eg. **r003**);
4. When “**r003**” flashes, confirm the deletion by pressing the **[OK]** key for a few seconds;
5. Release the **[OK]** key when appears “**r---**”;
6. The selected transmitter is deleted (display shows again P006).

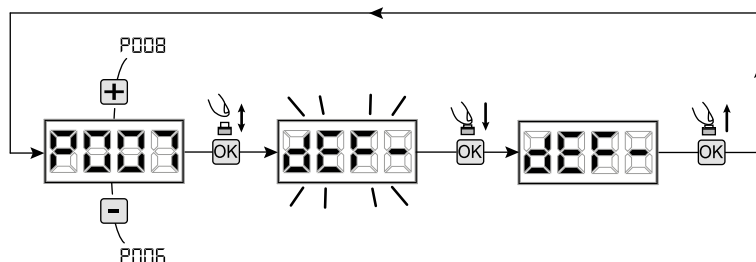


2 Restoring default parameters

2.1 Restoring operating parameters

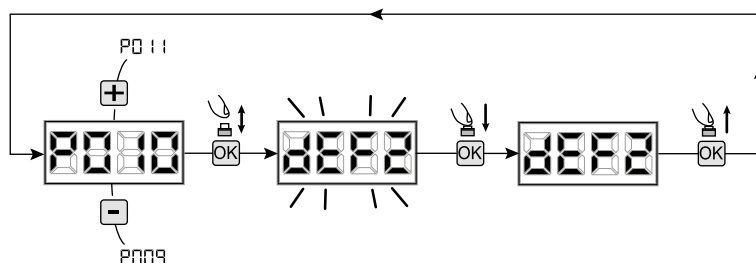
1. Scroll through the parameters with the buttons **[+]** and **[-]** until the display shows P007;
2. Confirm by pressing on the **[OK]** key;
3. When “**dEF1**” is flashing, press the **[OK]** key for a few seconds;
4. Release the **[OK]** key as soon as “**dEF1**” stops flashing; All the default values are restored except for the parameters from P016 to P022 and P076 to P098 for the configuration currently in use;
5. At the end of the operation display returns to P007.

Warning: After you restore the default parameters, you must program the control panel again and adjust all operating parameters, in particular, remember to properly set the operator configuration parameters. (P028 - P029 - P030).



2.2 Restoring “I/O” setting (Input/Output)

1. Scroll through the parameters with the buttons **[+]** and **[-]** until the display shows P010;
2. Confirm by pressing on the **[OK]** key;
3. When “**dEF2**” is flashing, press the **[OK]** key for a few seconds;
4. Release the **[OK]** key as soon as “**dEF2**” stops flashing; All the default values only for the parameters from P016 to P022 and from P076 to P098 are restored for the configuration currently in use;
5. At the end of the operation display returns to P010.

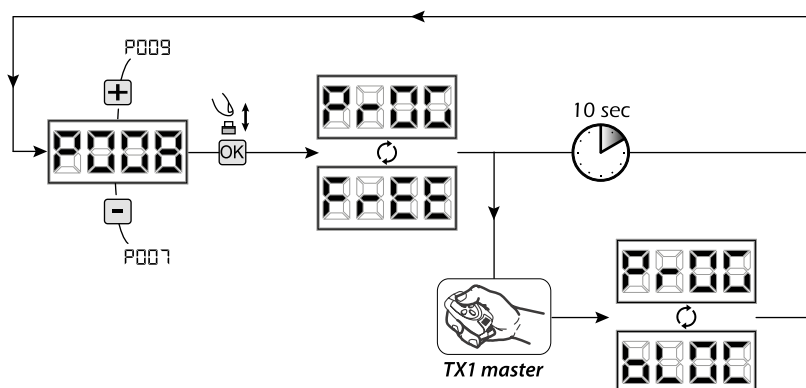


3 Locking-Unlocking access to programming

By using a “dip-switch” remote (regardless of the type of remotes already memorized) it's possible to lock-unlock access to the programming of the control panel to avoid tampering. The remote setting is the locking-unlocking code verified by the control board.

3.1 Locking access to programming

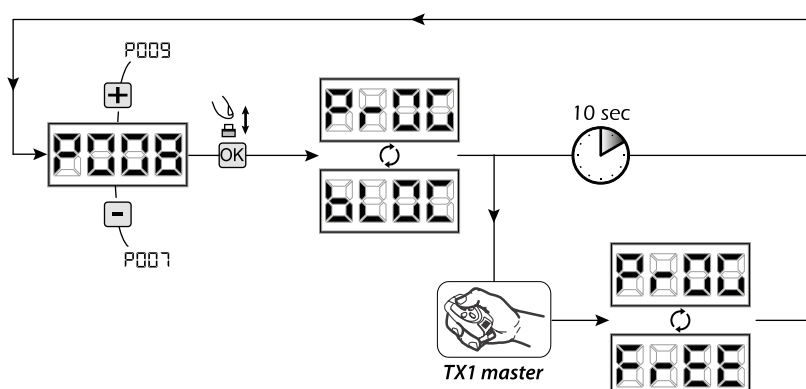
1. Scroll through the parameters with the buttons **[+]** and **[-]** until the display shows “P008”;
2. Access the parameter by pressing the button **[OK]**;
3. The display shows alternately the writing “**PrOG/ FrEE**” to indicate that the control board is waiting for the transmission of the block code;
4. Within 10 seconds press CH1 on the “TX Master”, the display shows “**PrOG/bLOC**” before returning to the list of parameters;
5. Access to programming is locked.



WARNING Programming lock/unlock can also be set via Smartphone using the DEAIInstaller APP. In this case, an installer code is set (other than zero) that can only be unlocked via APP.

3.2 Unlocking access to programming

1. Scroll through the parameters with the buttons **[+]** and **[-]** until the display shows “P008”;
2. Access the parameter by pressing the button **[OK]**;
3. The display shows alternately the writing “**PrOG/ bLOC**” to indicate that the control board is waiting for the transmission of the unlocking code;
4. Within 10 sec. press the CH1 of the “TX Master”, the display shows “**PrOG/ FrEE**” before returning to the list of parameters;
5. Access to programming is unlocked.



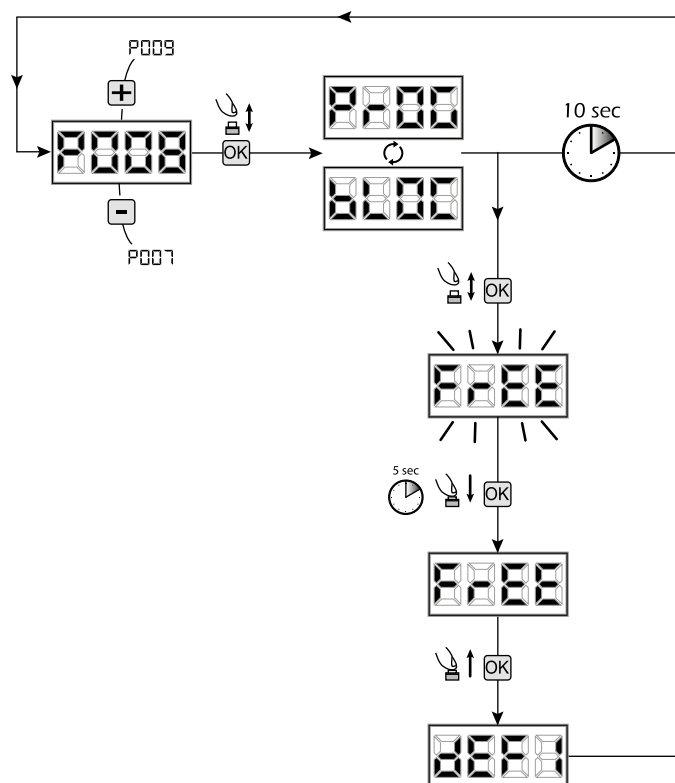
3.3 Unlocking access to programming and global reset

WARNING! This procedure involves the loss of all stored settings.

The procedure allows the unlocking of the control panel without having to know its unlocking code.

Following this release, you must program the control panel again and adjust all operating parameters, in particular, remember to properly set the configuration of parameters (P028 - P029 - P030 - operator configuration). You will also need to repeat the measurement of impact forces to ensure the installation compliance to standards.

1. Scroll through the parameters with the buttons **[+]** and **[-]** until the display shows “P008”;
2. Access the parameter by pressing the button **[OK]**;
3. The display shows alternately the writing “**PrOG/ bLOC**”;
4. Press the button **[OK]**, the display shows the flashing writing “**FrEE**”;
5. Press the button again and hold for 5 seconds (releasing it before, the procedure is terminated): The display shows the fixed writing “**FrEE**” followed by “**dEF1**”, before returning to the list of parameters;
6. Access to programming is unlocked.



4 Downloading/uploading data memory

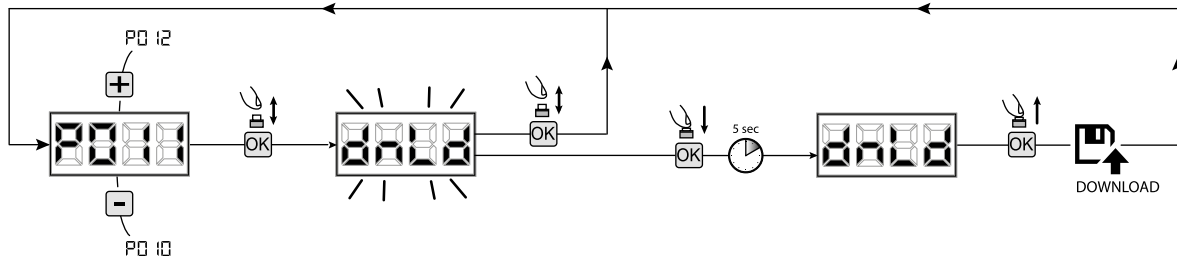
4.1 Downloading data to an external memory unit (DOWNLOAD)

1. Scroll down the parameters with **[+]** and **[-]** keys until you visualize P011;
2. Press the **[OK]** key, the display visualizes the word “dnLd” flashing;
3. Press the **[OK]** again and continue pressing it for 5 sec (if you release it before this period, the procedure is stopped);
4. Release the **[OK]** key as soon as the word “dnLd” stops flashing;

All the control panel configurations (TYPE, parameters, remotes, operators stroke, etc..) are saved in the external memory unit;

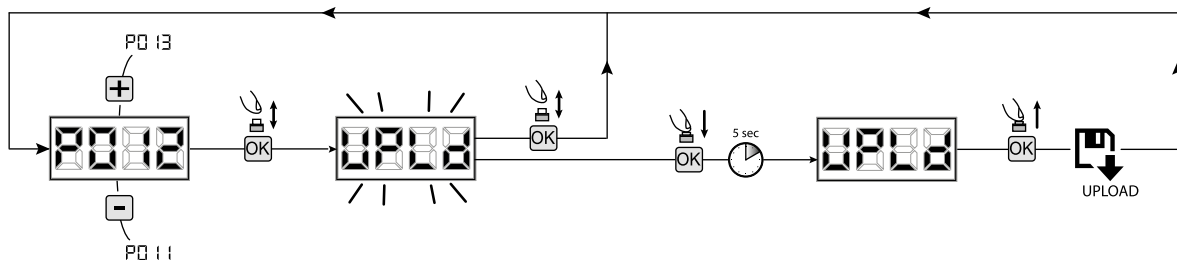
Warning: If there is any data in the external memory, during the memory download they will be overwritten.

5. At the end of the operation display returns to P011.



4.2 Uploading data from an external memory unit (UPLOAD)

1. Scroll down the parameters with **[+]** and **[-]** keys until you visualize P012;
 2. Press the **[OK]** key, the display visualizes the word “UPLd” flashing;
 3. Press the **[OK]** again and continue pressing for 5 sec (if you release it before this period, the procedure is stopped);
 4. Release the **[OK]** key as soon as the word “UPLd” stops flashing;
- All the control panel configurations (TYPE, parameters, remotes, operators stroke, etc..) contained in the external memory unit are uploaded in the connected control panel;
5. At the end of the operation display returns to P012.

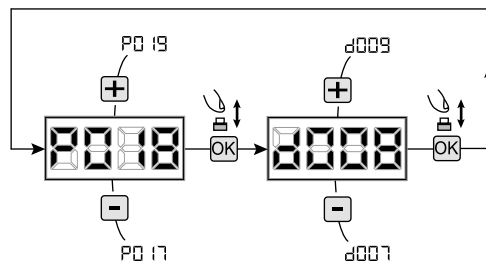


WARNING If you are not connected to any external storage units or if the connecting cable is disconnected during the data transfer operation, the display will visualize “Err9”, then the control unit is entirely reset and the display shows the word “TYPE” flashing. Refer to the instruction of the external memory card to restore the operation of the control panel.

5 Inputs configuration

Where the installation requires different commands and / or additional to the standard ones described by plan, you can configure each input for the operation desired (eg START, PHOTOS, STOP, etc ...).

1. Scroll down the parameters with the **[+]** and **[-]** to see that corresponding to the desired one:
 - P017=for INPUT 1;
 - P018=for INPUT 2;
 - P019=for INPUT 3;
 - P020=for INPUT 4;
 - P021=for INPUT 5;
 - P022=for INPUT 6;
2. Confirm by pressing on the **[OK]** key to get access to the parameter (eg. P018);
3. Scroll down with the **[+]** and **[-]**, keys to set the value corresponding to the desired operation (refer to table “Input Configuration parameters” on page 38);
4. Confirm by pressing on the **[OK]** key (display shows again P018).
5. Execute the new connection to the input just reconfigured.



6 Programming complete

WARNING At the end of the programming procedure, use the buttons **[+]** and **[-]** until the appearance of the symbol “---”, the operator is now ready again for new manoeuvres.

7 DESCRIPTION OF INPUTS

The table below provides a description of the operation of all selectable inputs on the board.

INPUTS (IN / EXP_IN)	
Message	Description
NONE	Not used
START	N.O. input start. In case of action, this results in the opening or closing. It may function in “inversion” (P049=0) or “step-by-step” (P049=1) modality.
PED	N.O. pedestrian start. In case of action, this results in the partial opening of the gate. The adjustment of the pedestrian pace can be set with P043.
OPEN	N.O. input open. In case of action, this results in the partial opening of the gate.
CLOSE	N.O. input close. In case of action, this results in the closing of the gate.
OPEN_PM	Dead man N.O. input open The gate opens for the amount of time in which the button is pressed.
CLOSE_PM	Dead man N.O. input close The gate closes for the amount of time in which the button is pressed.
ELOCK_IN	Electric lock activation N.O. input. In case of action, this leads to the activation of the “LOCK” output of the board, see P062.
PHOTO_1	Photoelectric cell 1 N.C. input. See P050 to select the operation mode. If this is not used, bridge out the connection.
PHOTO_2	Photoelectric cell 2 N.C. input. See P051 to select the operation mode. If this is not used, bridge out the connection.
SAFETY_1	Sensitive edge 1 N.C. input. See P067 to select the operation mode. If this is not used, bridge out the connection.
SAFETY_2	Sensitive edge 2 N.C. input. See P068 to select the operation mode. If this is not used, bridge out the connection.
STOP / SAS_INPUT	Stop N.C. input. In case of action, this stops the movement during any manoeuvre. If this is not used, bridge out the connection. N.C. contact. (SAS INPUT): If connected to WARN_FIX/SAS OUTPUT of a second control unit, causes the “bank door” to operate (disabling the opening of the second door until the first is not fully closed). Warning: Ensure that the output is free of voltage.
OPEN_INT (NET_EXP only)	Starts the operation and enables the green light ignition (arrival at open gate) only for the internal traffic light. Meanwhile, if the OPEN_EXT command is given, this books the next operation and the green light of the external traffic light turns on at the end of the Automatic Closing Time (ACT).
OPEN_EXT (NET_EXP only)	Starts the operation and enables the green light ignition (arrival at open gate) only for the external traffic light. Meanwhile, if the OPEN_INT command is given, this books the next operation and the green light of the internal traffic light turns on at the end of the Automatic Closing Time (ACT).
AUX_IN (NET_EXP only)	Input for the control of the AUX_OUT output.
FCA_1	Motor 1 opening end stop N.C. input If not used, disable the input with the corresponding parameter.
FCC_1	Motor 1 closing end stop N.C. input If not used, disable the input with the corresponding parameter.
FCA_2	Motor 2 opening end stop N.C. input If not used, disable the input with the corresponding parameter.
FCC_2	Motor 2 closing end stop N.C. input If not used, disable the input with the corresponding parameter.
SAFETY_INHIBITION	N.C. input SAFETY inhibition. When open, it causes the by-pass of the SAFETY inputs, which are ignored even if activated.
RESET	N.C. Contact for micro-release connection; opening the contact triggers a central reset

8 MESSAGES SHOWN ON THE DISPLAY

WORKING STATUS MESSAGES		
Mess.	Description	
----	Gate is closed	
_ _ _ _	Gate is opened	
OPEN	Opening under way	
CLOS	Closing under way	
STEP	While in step-by-step mode, the control board awaits further instructions after a start command	
STOP	Stop input intervened or an obstacle is detected with limited inversion duration (P055 > 0 or P056 > 0)	
L L	Board in BOOT-MODE: Indicates that the firmware is corrupted or updating. To restore the firmware, use the DEAIstaller APP and make sure NET-NODE is corrected to the correct port. Warning: When updating the firmware, all data in the board memory (settings and radio commands) are lost. Make sure you have backed up the memory to be restore data after the update.	
RESP	Reset current position: The control unit has just been turned on after a power failure, or the gate has exceeded the maximum number (80) of inversions allowed without ever getting to the closing stroke, or the maximum number (15) of consecutive operations allowed of the anti-crushing device. Once the control unit has been reset and open command given the gate will start moving at slow speed, until it reaches end of travel.	
ERROR MESSAGES		
Mess.	Description	Possible solutions
ErrP	Error position: The reset position procedure is not successful. The control panel is awaiting commands.	- Make sure there are no specific frictions and / or obstacles during the run; - Give a start pulse to initiate a position reset procedure; - Verify that the operation is completed successfully, manually helping the run, if necessary; - Adjust power and speed settings if necessary.
BLCK UPRT	Board programming attempted when a NET-NODE device is connected.	Turn off power, disconnect the NET-NODE from the communication port and turn back on;
Err3	External photocells and/or safety devices are activated or out of order.	- Make sure that all safety devices and/or photocells installed are working properly.
Err4	Possible fault/overheating in the control unit's power circuit.	Turn off power for several minutes and turn back on. Give a start command: if the message is repeated, replace the control unit.
Err5	Time-out operators run: The engine/s exceeded the maximum operating time (4min) without ever stopping.	- Give a start pulse to start the position reset procedure; - Ensure that this operation is successful.
Err6	Time-out obstacle detection: With anti-crushing sensor disabled, was still detected the presence of an obstacle that prevents movement of the leaf for a period of 10 seconds more.	- Make sure there are no specific frictions and / or obstacles during the run; - Give a start pulse to initiate a position reset procedure; - Verify that the operation is completed successfully.
Err7	Operators mouvement not detected.	- Make sure that operators and encoders connections are well done. - Check that jumpers J5 and J9 are well positioned as shown on the electric wiring. - If this error appears again, replace the control panel.
Err8	The power consumption of a device connected to a 24V output exceeds safety limits. Internal malfunction in the control panel.	- Ignore the message if the error briefly appears on the display when the power to the control panel is removed. - Disconnect all auxiliary devices; if the error disappears, reconnect the devices one by one until you identify the one causing the overload. - If the error persists, replace the control panel.
Err9	No/interrupted communication with remote memory board (also NET-EXP or NET-NODE).	- Check that the connecting cable of the external memory card is connected properly. - If you are performing a data transfer operation (DOWNLOAD / UPLOAD), make sure that it is not interrupted (eg by unplugging the card before the end of the operation). Please note: the interruption of an UPLOAD, also involves a total RESET of the control unit.
Err10 Err11	Possible fault/overheating in the control unit's power circuit.	Turn off power for several minutes and turn back on. Give a start command: if the message is repeated, replace the control unit.
Err12	Possible malfunction in the control unit's power circuit or in the encoder circuit.	Check the wiring of the encoder and the motor. Shut the power supply off and on again. Give a start command: if the message is repeated, perform the following checks. - Enter P003 and move the door using the + and - buttons. - If the door moves at maximum speed and the display shows Err7, replace the motor's encoder card. - If the motor still remains stationary, replace the control unit.
Err15	Sensitive regulation parameters were edited via DEAIstaller APP without running motor stroke learning at the end of the operation.	Run motor stroke learning (P003) first to be able to run any other operation.
ErrB1	NET-NODE connected to the incorrect communication port.	Connect NET-NODE to the correct port according to that indicated in the control unit diagram.

9 DETAILED PARAMETER LIST

Programming Procedures

P001	Positioning of operator 1
P002	Positioning of operator 2
P003	Memorization of the motors' stroke
P004	Deletion of transmitters
P005	Transmitters memorizing
P006	Search and deletion of a transmitter
P007	Restoring the operating parameters
P008	Lock access to programming
P009	How to learn connected DE@NET devices (unused at the moment)
P010	Restoring the "I/O" configurations (input/output)
P011	Downloading data on the external memory unit
P012	Uploading data from an external memory unit
P013	Visualisation of inputs and operations-counter status
P014	Unused parameter
P015	Unused parameter

Input Configuration Parameters

Input Configuration Parameters			Default TYPE			
			00	01	02	03
P016	INPUT_3 selectioning input type		000	000	000	000
	• 000: IN3 type=free contact • 001: IN3 type=constant resistance 8K2					
P017	INPUT_1		001	001	001	001
P018	INPUT_2		002	002	008	008
P019	INPUT_3		010	010	010	000
P020	INPUT_4		008	008	011	000
P021	INPUT_5		012	009	000	000
P022	INPUT_6		014	011	000	000
	• 000: NONE • 001: START • 002: PED • 003: OPEN • 004: CLOSE • 005: OPEN_PM • 006: CLOSE_PM • 007: ELOCK_IN	• 008: PHOTO_1 • 009: PHOTO_2 • 010: SAFETY_1 • 011: STOP • 012: FCA_1 • 013: FCA_2 • 014: FCC_1 • 015: FCC_2	• 016: SAFETY_2 • 017: OPEN_INT • 018: OPEN_EXT • 019: AUX_IN • 020: SAFETY_INHIBITION • 021: RESET			
P023	Remote control button 1		001	001	001	001
P024	Remote control button 2		000	000	000	000
P025	Remote control button 3		000	000	000	000
P026	Remote control button 4		000	000	000	000
	• 000: NONE • 001: START • 002: PED • 003: OPEN	• 004: CLOSE • 005: Not Used • 006: Not Used • 007: ELOCK_IN	• 008: AUX_IN • 009: STOP			
P027	Radio coding		000	000	000	000
	• 000: HCS FIXED CODE • 001: HCS ROLLING CODE	• 002: DIP SWITCH (HT12) • 003: DART				
Warning: if the type of encoding must be varied and only if the memory already contains remote controls with different codes, the memory deletion procedure (P004) must be carried out AFTER setting the new code.						

Motor Configuration Parameters

		Default TYPE			
		00	01	02	03
P028	Selection type of operators	005	005	003	003
	TYPE 00				
	• 005: LIVI_6/24 - 6/24 BOOST • 006: LIVI_9/24	• 007: REV24 • 008: REV24 BOOST			
	TYPE 01				
	• 000: GEKO • 001: LOOK - MAC - STING	• 002: GHOST 100 - GHOST 200 • 003: LIVI 502/24 - ANGOLO • 004: LIVI 502MT/24 • 005: GEKO/X			
	TYPE 02				
	• 003: LIVI 902/24 - 905/24	• 004: LIVI 902R/24			
	TYPE 03				
	• 003: PASS 24_N	• 004: STOP 24_N			
P029	Selected work with or without encoders	001	001	000	000
	WARNING: Remember to correctly set the jumpers J5 and J9 (see board terminal table) WARNING: J5, J9 and P029 must be set correctly before performing the procedure for programming	• 000: motors with encoder • 001: engines without encoder			
P030	Selectioning operators number	001	002	001	001
	• 001: one operator • 002: two operators				

EN

Operating Parameters

		Default TYPE			
		00	01	02	03
P031	Operators speed adjustment during slow-down while opening	040	050	050	030
		15%.....100%			
P032	Operators speed adjustment during the stroke while opening	100	100	100	100
		15%.....100%			
P033	Operators speed adjustment during the stroke while closing	100	100	100	100
		15%.....100%			
P034	Operators speed adjustment during slow-down while closing	040	050	050	030
		15%.....100%			
P035	Slow down duration adjustment while opening	025	020	020	030
		0%.....80%			
P036	Slow down duration adjustment while closing	025	020	020	030
		0%.....80%			
P037	Operator 1 force adjustment while opening	050	050	050	099
	if = 100% obstacle detection deactivated	15%.....100%			
P038	Operator 1 force adjustment while closing	050	050	050	099
	if = 100% obstacle detection deactivated	15%.....100%			
P039	Operator 2 force adjustment while opening	050	050	000	099
	if = 100% obstacle detection deactivated	15%.....100%			
	TYPE 02 ONLY: Secondary force adjustment in closing: adjusts the motor's force during the last part of the closing movement defined by P058.	0%.....100%			
P040	Operator 2 force adjustment while closing	050	050	000	099
	if = 100% obstacle detection deactivated	15%.....100%			
P041	Automatic closing times adjustment	000	000	000	000
	if = 0 automatic closing deactivated	0s.....255s			
P042	Pedestrian automatic closing time adjustment	000	000	000	000
	if = 0 pedestrian automatic closing deactivated	0s.....255s			
P043	Pedestrian stroke duration adjustment	030	035	035	100
		5%.....100%			

P044	Pre-flashing time adjustment	000	000	000	000
		0s.....10s			
P045	Adjustment of phase displacement time while opening	/	001	/	/
		0s.....30s			
P046	Adjustment of phase displacement time while closing	/	003	/	/
		0s.....30s			
P047	Condominium function	000	000	000	001
	If it is activated it deactivates both opening and closing inputs for the whole duration of automatic opening and closing	• 000: disabled • 001: activated only upon opening • 002: activated on automatic opening and closing			
P048	Ram blow function	000	000	000	000
	If=0 "Ram blow" function deactivated; if=1 it pushes the motors closed for one second before each opening movement, so as to ease the releasing of any electric lock; if>1 it execute a periodic pushing stroke so as to maintain the wings under pressure on the closing strokes. If closing limit switches are installed, it performs this function only if they are not activated, i.g. when there's a pressure decrease on the stroke.	• 000: "ram blow" deactivated • 001: "ram blow function" activated • >001: "ram blow" periodic (X*1 min) (2.....255)			
P049	"STEP-BY-STEP" enabling	001	001	001	000
	"Reversal" mode selection (during the manoeuvre a command impulse reverse the mouvement) or "step by step" (during the manoeuvre a command impulse stops the mouvement). A next impulse restart the operator to the opposite direction.	• 000: "reversal function" • 001: "step by step function"			
P050	PHOTO_1	002	002	002	002
	If=0: photocell enabled while closing and starting when the gate is stopped; if=1 photocells are always enabled; if=2 photocells are enabled while closing only. When enabled, its activation provokes: the inversion (while closing), the stop (while opening) and prevent the starting (when gate is closed). If = 3-4-5, the operation is identical to values 0-1-2 but with the "close immediately" function enabled: in any case, when opening and / or pausing time, removing a any obstacle the gate will end the opening maneuver before closing it automatically after a 1 second fixed delay.	• 000: photocell enabled while closing and when gate is stopped • 001: photocells always enabled • 002: photocells enabled only while closing • 003: as 000 but with "close immediately" enabled • 004: as 001 but with "close immediately" enabled • 005: As 002 but with "close immediately" enabled			
P051	PHOTO_2	000	001	002	002
	If=0: photocell enabled while closing and starting when the gate is stopped; if=1 photocells are always enabled; if=2 photocells are enabled while closing only. When enabled, its activation provokes: the inversion (while closing), the stop (while opening) and prevent the starting (when gate is closed). If = 3-4-5, the operation is identical to values 0-1-2 but with the "close immediately" function enabled: in any case, when opening and / or pausing time, removing a any obstacle the gate will end the opening maneuver before closing it automatically after a 1 second fixed delay.	• 000: photocell enabled while closing and when gate is stopped • 001: photocells always enabled • 002: photocells enabled only while closing • 003: as 000 but with "close immediately" enabled • 004: as 001 but with "close immediately" enabled • 005: As 002 but with "close immediately" enabled			
P052	Operation mode selection of the WARNING LIGHT output	000	000	060	000
	If = 0 "warning light" (output always ON when the gate is open, OFF after a closing operation), If = 1 "flashing warning light" (slow intermittent output during opening and fast while closing, always ON at gate opened, always OFF at the end of a closing operation only), If> 1 "courtesy light" (output ON during each movement, OFF when the motor stops, after the setting delay)	• 000: "fix warning light" • 001: "flashing warning light" • >001 : "courtesy light" off delay (2sec.....255sec)			
P053	RESP and stop management	/	000	001	001
	When activated, operators stop only at their arrival et the end of stroke, also while opening. Warning: During the emergency operation (RESP), the motor executes the first maneuver while opening. In addition, if any limit switches, the parameter is forced to 1.	• 000: Stop when opening on a memorized point • 001: Stop when opening on the end of stroke			
P054	"SOFT START" function	001	001	001	001
	Motors accelerate gradually until they reach the set speed, avoiding sudden departures ONLY TYPE 02: If=3 the opening slow space (P035) also becomes the space within which the port moves at slow speed (P031) and close start.	• 000: "soft start" deactivated • 001: "soft start" activated • 002: "long soft start" activated • 003: "settable soft start" on (TYPE 02 only)			

P055	Reversal due to obstacle during opening	003	003	003	000
	Adjust the inversion on obstacle period (detected by internal anti-crushing sensor or by the safety input when activated): If = 0 it makes a complete inversion, if > 0 indicates the duration (in seconds) of the run, after inversion resulting from detection of an obstacle during the opening.	• 000: complete reversal on obstacle • >000: duration of reversal on obstacle (1sec.....10sec)			
P056	Reversal due to obstacle during closing	003	003	003	000
	Adjust the inversion on obstacle period (detected by internal anti-crushing sensor or by the safety input when activated): If = 0 it makes a complete inversion, if > 0 indicates the duration (in seconds) of the run, after inversion resulting from detection of an obstacle during the closing.	• 000: complete reversal on obstacle • >000: duration of reversal on obstacle (1sec.....10sec)			
P057	Manual unlocking facilitation with gate closed	000	001	003	002
	If≠0, after ending the closing or opening maneuver, the engine reverses for a brief time to release the pressure on it, and thus facilitate the manual release. The set value shows the length of the inversion. If=0 function disabled	• 000: facilitazione sblocco disattivata • >000: facilitazione sblocco attivata con durata pari a: (1x25ms.....20x25ms) (1x25ms.....40x25ms) (TYPE 00 only)			
P058	Adjustment of the opening stroke margin	012	025	000	020
	It adjusts the duration of the last part of the stroke during which an obstacle is interpreted as a stroke, blocking the motor without performing the inversion. For motors with encoders, the set value indicates the number of revolutions of the rotor; while for motors without encoder, the value is expressed in% of the maximum stroke. Warning: for motors without encoder, if P035 (duration slow-down while opening) is >10%, it forces the stroke detection margin so that it's the same than the slow-down.	1.....255 (motors with encoder) 1%.....100% (motors without encoder)			
	TYPE 02 ONLY: Duration adjustment for the secondary force in closing: adjusts the duration of the last part of the closing movement, in which the force is managed separately with P039. The value is expressed in number of revolutions of the rotor.	0.....255			
P059	Adjustment of the closing stroke margin	012	025	025	020
	It adjusts the duration of the last part of the stroke during which an obstacle is interpreted as a stroke, blocking the motor without performing the inversion. For motors with encoders, the set value indicates the number of revolutions of the rotor; while for motors without encoder, the value is expressed in% of the maximum stroke. Warning: for motors without encoder, if P036 (duration slow-down while closing) is >10%, it forces the stroke detection margin so that it's the same than the slow-down.	1.....255 (motors with encoder) 1%.....100% (motors without encoder)			
	TYPE 02 ONLY: Adjustment of the stop-margin in closing: adjusts the duration of the last part of the closing movement, in which an obstacle is seen as a stop, causing the motor to stop without reversal on the obstacle. The value is expressed in number of revolutions of the rotor.	1.....255			
P060	Operators force adjustment at stroke arrival	000	035	000	000
	Operators force adjustment at stroke arrival - If=0, setting off (the force value on the stroke is calculated automatically) - If≠0 (operators with encoder) it indicates the force value (expressed in% of the max value) set in the last length. TYPE 02 ONLY: Force adjustment in the stop margin in closing, its duration is set through P059.	0%.....100%			
P061	"ENERGY SAVING" function	000	000	000	000
	If=1 after 10sec of inactivity, the control panel turns the 24V outputs and the display off that will be turned on at first command received (use recommended battery-powered and / or solar panel). Warning: when "Energy saving" is enabled, SAS function is not available. Warning: when "Energy saving" is enabled, only the stabilized output 24V_ST must be used to power accessories.	• 000: "Energy saving" not active • 001: "Energy saving" active			

P062	Electric lock output activation	000	000	000	005
	If=0 "boost" output for electric-lock art.110 power supply, If=1 24V output controlled by the ELOCK_IN input as pulsed mode, If=2 24V output controlled by the ELOCK_IN input as step-by-step mode, If=3 electro-brake output for not self-locking operators, If=4 24V output for electric-lock power supply via an external relay, If=5 24V output for electro-magnets power supply for barriers, If>5 24V output controlled by the ELOCK_IN input as temporized mode (the set value indicates the switch-off delay in seconds). Warning: To adjust the activation/deactivation times in the 000 004 005 modes, use parameter P064.	• 000: "Boost" output for electric-lock art.110 power supply • 001: "24V === pulse output max 5W • 002: "24V === step-by-step output max 5W • 003: "Electro-brake output for not self-locking operators • 004: "Output for electric-lock power supply via an external relay • 005: "output for electro-magnets power supply for barriers • >005: "24V === temporized output max 5W (6sec.....255sec)			
P063	Motor running direction	000	000	000	000
	If=1 automatically reverses the outputs open/close of the operators, avoiding having to manual change the wiring when installing the operator in an inverted position. Warning: Changing this parameter you need to change the parameters for the opening and closing limit switches.	• 000: "Standard installation" • 001: "Inverted installation"			
P064	Electric lock duration adjustment	002	002	002	002
	If P062=000 004, adjust the activation time of the LOCK output; If P062=005, adjust the deactivation time of the LOCK output;	0s.....10s			
P065	Maintenance manoeuvre counter	000	000	000	000
	If = 0 reset the counter and disables the intervention request , if> 0 indicates the number of operations (x 500) to be made before the control panel executes a 4 second additional pre-flash to indicate the need of maintenance. i.g.: If P065 = 050, operations number = 50x500 = 25000 operations Warning: Before you set a new value of the counter-manoevres maintenance, the same must be reset by setting P065= 0 and only later P065 = "new value".	• 000: "Request Maintenance disabled • >000: "Number of operations (x 500) for required maintenance (1.....255)			
P066	Selection of operating flashing light output	001	001	001	001
	If=0 intermittent flashing light output; If=1 Fixed flashing light output (for flashing lights with intermittent interior circuits).	• 000: "intermittent flashing light output • 001: "fixed flashing light output			
P067	SAFETY_1	000	000	000	000
	If = 0 safety edge always enabled, if = 1 safety edge enabled only while closing, if = 2 safety edge enabled only while closing and before any movement, if = 3 safety edge enabled only when opening, if = 4 safety edge enabled only while opening and before any movement; as for the obstacle detection with internal anti-crushing sensor, also the activation of the inputs SAFETY_1 and SAFETY_2 causes the complete or partial reversal as set by P055 (duration of inversion on obstacles while opening, and P056 (duration of reversal on obstacle while closing)	• 000: "safety edge always enabled • 001: "safety edge enabled only while closing • 002: "safety edge enabled only while closing and before any movement • 003: "safety edge enabled only when opening • 004: "safety edge enabled only while opening and before any movement			
P068	SAFETY_2	000	000	000	000
	If = 0 safety edge always enabled, if = 1 safety edge enabled only while closing, if = 2 safety edge enabled only while closing and before any movement, if = 3 safety edge enabled only when opening, if = 4 safety edge enabled only while opening and before any movement; as for the obstacle detection with internal anti-crushing sensor, also the activation of the inputs SAFETY_1 and SAFETY_2 causes the complete or partial reversal as set by P055 (duration of inversion on obstacles while opening, and P056 (duration of reversal on obstacle while closing)	• 000: "safety edge always enabled • 001: "safety edge enabled only while closing • 002: "safety edge enabled only while closing and before any movement • 003: "safety edge enabled only when opening • 004: "safety edge enabled only while opening and before any movement			
P069	Delay on limit switch detection	000	000	000	000
	The operation is stopped after 1,5 sec from limit switch detection. When during this delay a stop is detected, the operator is suddenly stopped	• 000: "limit switch delay disabled • 001: "limit switch delay enabled			
P070	Adjustment of acceleration durability	200	200	200	200
	Warning: if soft start is activated, the acceleration is deactivated indipendently from P070 value.	• 000: "acceleration deactivated (it runs an acceleration of minimum durability, almost imperceptible) • 00X: "adjusts the acceleration durability at 1,5 sec (X*6 ms)			

P071	Safeties self-test	000	000	000	000
	If = 0 24V === output with autotest disabled; if = 1 24V === output for safeties with self-test (it turn the output off and check the contact opening before each maneuver). Attention: In order to work in self-test mode, all devices must be connected to the stabilized output 24V_ST (1-2), and be wired and aligned before the motor stroke learning (P003).	• 000: “net power supply (safeties self-test disabled) • 001: “safeties self-test enabled			
P072	“SAS” function (NET_EXP only)	000	000	000	000
	SAS output is connected to an input STOP / SAS INPUT of a second control panel, causing the operation “trap man” (disabling the opening of the second door as long as the first is not completely closed). If this parameter is enabled after a reset, it performs an automatic RESP during which the SAS output is not activated. If limit switches are present and they are crushed after a reset, the RESP is not executed. Warning: if both doors are manually unlocked and moved from the closed position creates the interlock condition. You will then need to manually close at least one of the two doors.	• 000: “SAS function” deactivated • 001: “SAS function” activated			
P073	Forced “Hold to Run”	000	000	000	000
	If this function is enabled, all inputs configured as OPEN and CLOSE change automatically also to OPEN_PM and CLOSE_PM (hold-to-run commands) if activated and kept active in case a safety contact (photocell and/or safety edge) is triggered. This function thus allows to control the automation even in case the safety devices are faulty. If the input is no longer maintained active, the automation returns to automatic operation. When using safety edges configured as SAFETY_1 or SAFETY_2, this function is not compatible with the values 001 and 003 of parameters P067 and P068. <u>For security reasons, we recommend that you DO NOT use this function in case there are any clocks/timers connected to the inputs configured as OPEN or CLOSE.</u>	• 000: function disabled • 001: function enabled (forces switch to Hold-to-run mode when safeties are triggered and OPEN/CLOSE commands are maintained)			
P074	Unused parameter				
P075	Unused parameter				
P076	Unused parameter				
P077	Unused parameter				
P078 ... P099	Configuration parameters dedicated to the expansion card NET_EXP (for a detailed description of the parameters, refer to the instruction manual).				

10 INSTALLATION TEST

The testing operation is essential in order to verify the correct installation of the system. **DEA System** wants to summarize the proper testing of all the automation in 4 easy steps:

- Make sure that you comply strictly as described in paragraph "WARNINGS SUMMARY";
- Test the opening and closing making sure that the movement of the leaf match as expected. We suggest in this regard to perform various tests to assess the smoothness of the gate and defects in assembly or adjustment;
- Ensure that all safety devices connected work properly;
- Perform the measurement of impact forces in accordance with the standard EN12453 to find the setting that ensures compliance with the limits set by the standard EN12453.

11 PRODUCT DISPOSAL

DISASSEMBLY

The automation unit must be dismantled by qualified personnel, in accordance with the current accident prevention and safety regulations, and with reference to the installation instructions, but in the reverse order. Before initiating the disassembly operations, disconnect the electrical power and make sure it cannot be reconnected.

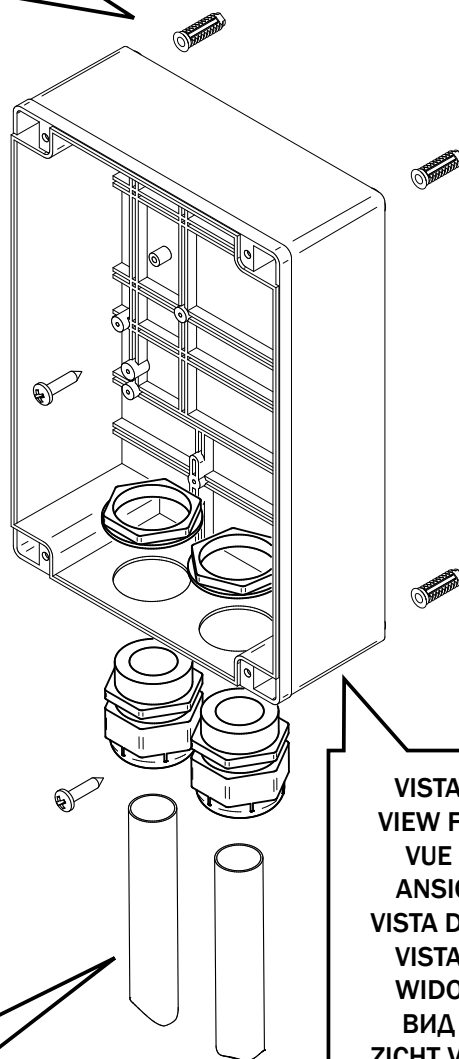
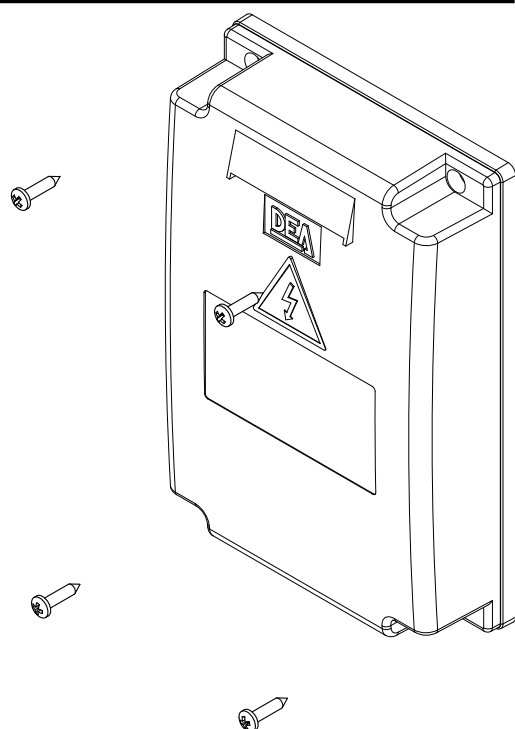
DISPOSAL

The automation unit must be disposed of in accordance with the current local and national waste disposal regulations. The product (or its individual parts) must not be disposed of together with other household waste materials.



WARNING In compliance with EU Directive 2012/19/EU on waste electrical and electronic equipment (WEEE), this electrical product should not be treated as municipal mixed waste. Please dispose of the product and bring it to the collection for an appropriate local municipal recycling.

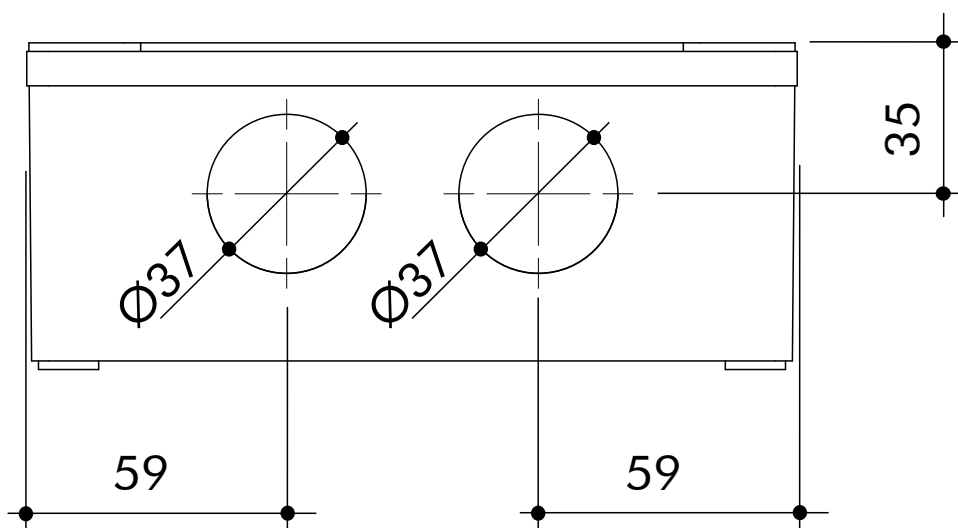
Eseguire il fissaggio alla parete usando opportuni tasselli per viti Ø5 (non fornite); **Fix** the box on the wall with appropriate bushings to anchor screws Ø5 (not included); **Le** fixer au mur en utilisant des douilles à expansion pour vis adéquates Ø5 (pas incluses); **Die** Wandbefestigung vornehmen, verwenden Sie geeignete Dübel für Ø5 Schrauben (nicht im Lieferumfang); **Efectuar** la fijación a la pared utilizando adecuados tacos para tornillos de Ø5 (no incluidos); **Executar** a fixação a parede usando apropriadas rolhas para parafusos Ø5 (não fornecidas); **Zamocować** do ściany, przy pomocy odpowiednich kołków do śrub Ø5 (nie na wyposażeniu); **Выполнить** крепление к стене с помощью соответствующих дюбелей для болтов Ø 5 (не входят в комплект); **Bevestig** het aan de muur met geschikte pluggen voor schroeven met een diameter van 5 mm (niet meegeleverd).



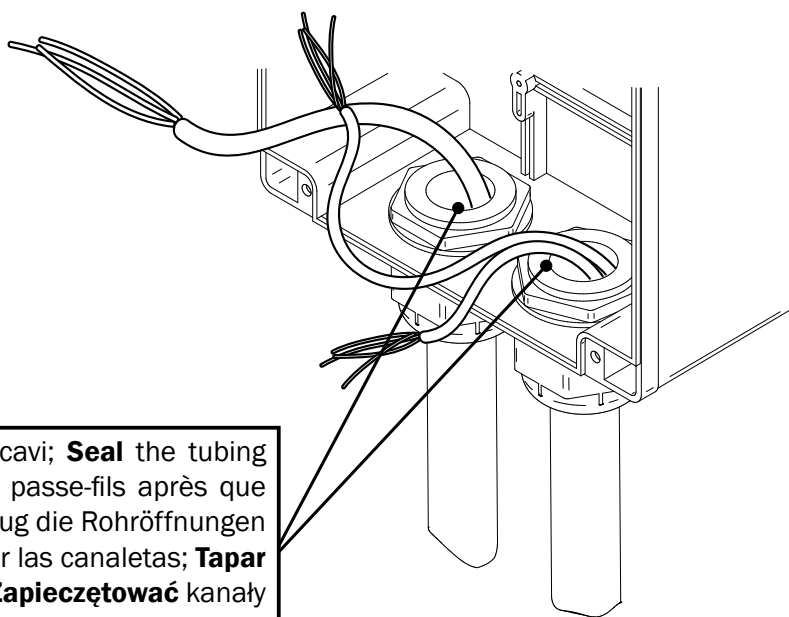
VISTA DA "A"
VIEW FROM "A"
VUE DE "A"
ANSICHT "A"
VISTA DESDE "A"
VISTA DE "A"
WIDOK Z "A"
ВІД ИЗ "A"
ZICHT VANAF "A"

Passaggio cavi 230V~ all'interno di una canaletta Ø20 raccordata con fermatubi PG29 (non forniti); **Pass** 230V~ cables inside a grommet Ø20 connected with tube fastening PG29 (items not included); **Passage** des fils 230V~ dans un passe-fil Ø20 raccordée avec un bloque tube PG29 (ces outils ne sont pas inclus); Kabelführung für die 230V~ Einspeisung in Ø20 Kunststoffrohr mit PG29 Rohrverschraubung (nicht im Lieferumfang); **Paso** de los cables 230V~ por el interior de una canaletta de Ø20 unida con pasacable PG29 (no incluidos); **Passagem** cabos 230V~ ao interno de um cano Ø20 com fixação do tubo PG29 (não fornecidos); **Przejście** kabli 230V~ wewnątrz kanaliku Ø20 połączonego z zaciskami przewodów PG29 (nie na wyposażeniu); **Прход** кабелей 230V~ внутри канала Ø20, связанного с фиксаторами PG29 (не входят в комплект); **Voer** 230V~ kabels door een Ø20 kanaal dat is verbonden met PG29 kabelwartels (niet meegeleverd).

Passaggio cavi a bassissima tensione all'interno di una canaletta Ø20 raccordata con fermatubi PG29 (non forniti); **Pass** very low tension cables inside a grommet Ø20 connected with tube fastening PG29 (items not included); **Passage** des fils à très basse tension dans un passe-fil Ø20 raccordée avec un bloque tube PG29 (ces outils ne sont pas inclus); Kabelführung für die Schwachstromkabel in Ø20 Kunststoffrohr mit PG29 Rohrverschraubung (nicht im Lieferumfang); **Paso** de los cables de tensión muy baja por el interior de una canaletta de Ø20 unida con paratubo PG29 (no incluidos); **Passagem** cabos a baixíssima tensão ao interno de um cano Ø20 com fixação do tubo PG29 (não fornecidos); **Przejście** kabli bardzo niskiego napięcia wewnątrz kanaliku Ø20 połączonego z zaciskami przewodów PG29 (nie na wyposażeniu); **Прход** кабелей очень низкого напряжения внутри канала Ø20, связанного с фиксаторами PG29 (не входят в комплект); **Voer** laagspanningskabels door een Ø20 kanaal dat is verbonden met PG29 kabelwartels (niet meegeleverd).



VISTA DA "A" Fori da eseguire sul fondo della scatola con seghe a tazza Ø37 per l'inserimento dei fermatubi; **VIEW FROM "A"** Holes to be drilled on the bottom of the box with a hole saw Ø37 to introduce tube fastening; **VUE DE "A"** Trous à percer au fond du boîtier avec une scie-cloche Ø37 afin d'introduire des bloque tube; **ANSICHT "A"** Mit einem 37mm Kronenbohrer die Rohrdurchführungen vohrnehmen; **VISTA DESDE "A"** Agujeros que deben hacerse en la base de la caja con sierras cilíndricas de Ø37 para la introducción de los paratubo; **VISTA DE "A"** Furos pra executar no fundo da caixa com serra a xicara Ø37 para inserimento dos fixação do tubo; **WIDOK Z "A"** Otwory do wykonania na dnie skrzynki z wiertłami Ø37 dla włożenia zacisku; **ВИД ИЗ "A"** Отверстия для выполнения в нижнем основании ящика с помощью кольцевой пилы Ø37 для установки фиксаторов; **ZICHT VANAF "A"** Boor gaten in de bodem van de behuizing met een Ø37 gatzaag voor het invoegen van de kabelwartels.



Sigillare le canalette dopo il passaggio dei cavi; **Seal** the tubing trays after installing the wires; **Étanchez** les passe-fils après que vous avez passé des fils; **Nach** dem Kabeleinzug die Rohröffnungen abdichten; **Una** vez colocados los cables, tapar las canaletas; **Tapar** os cabos depois de passar os fios eléctricos; **Zapieczerować** kanały po przejściu kabli; **Плотно** закрыть каналы после выполнения прохода кабелей; **Dicht** de kanalen af nadat de kabels zijn doorgevoerd.

NET24N/C

